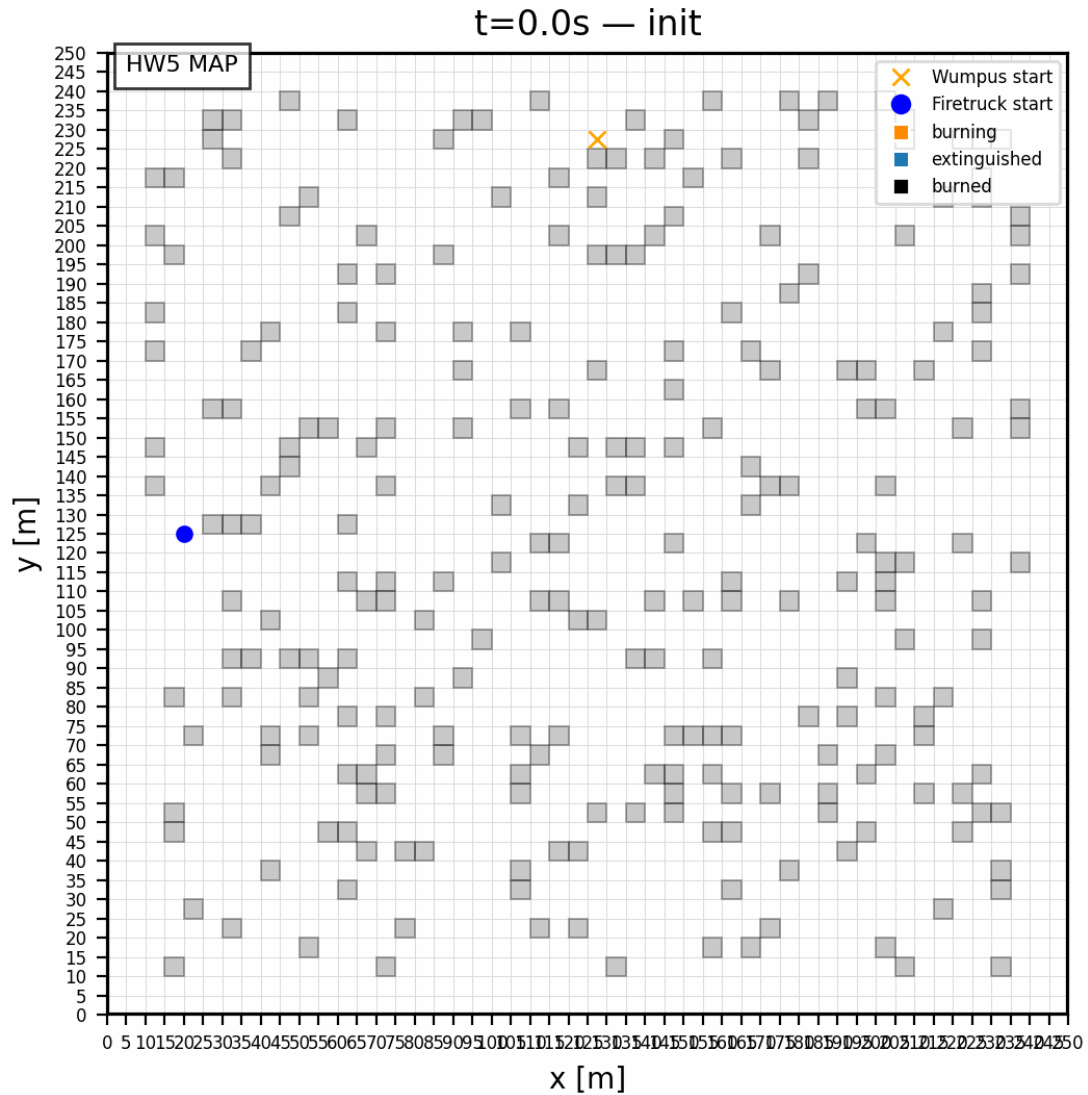


This project simulates a wildfire scenario using two planners:

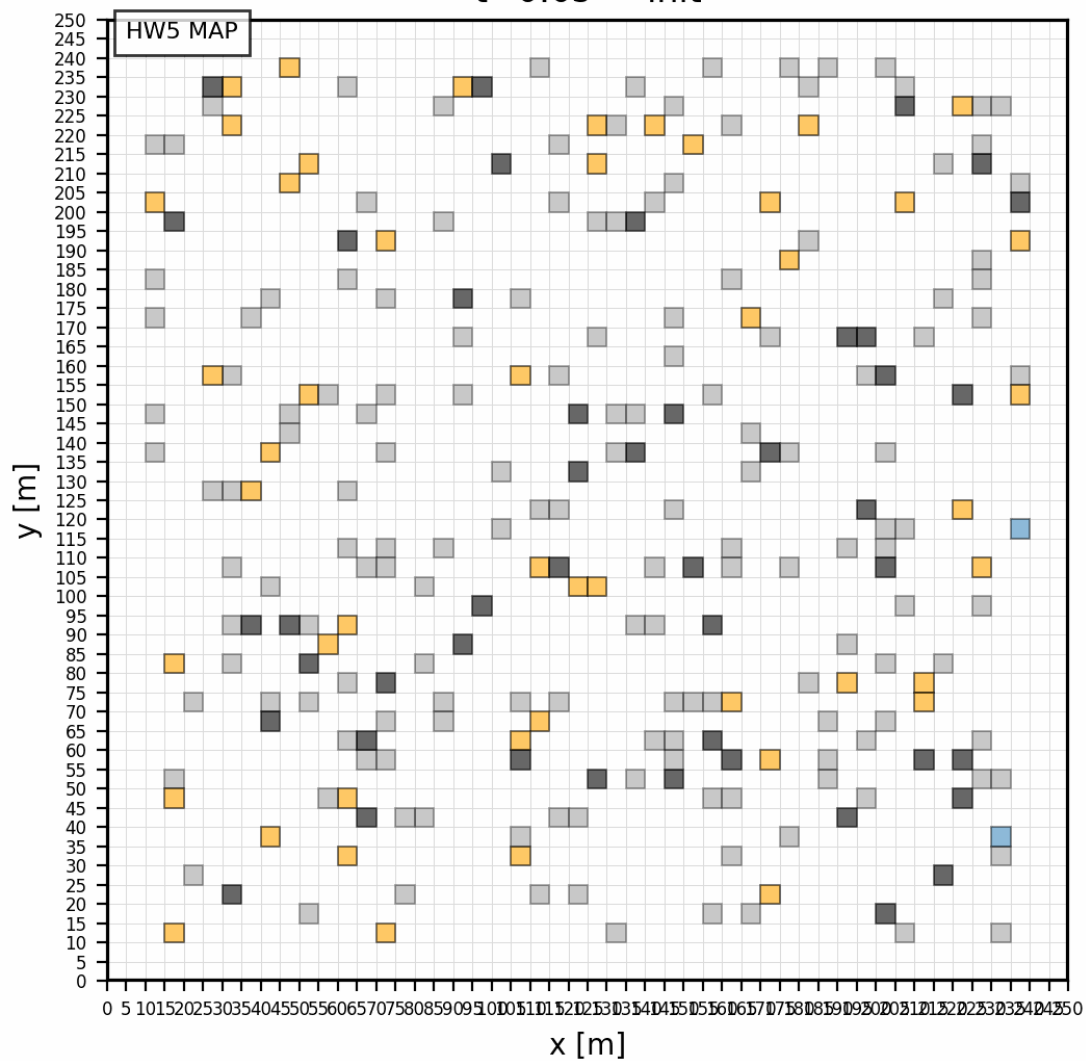
- **Wumpus (A*)** – a discrete, grid-based arsonist that lights fires in nearby cells.
- **Firetruck (PRM)** – a sampling-based Ackermann vehicle that navigates the map and extinguishes burning obstacles.

Each agent operates in a **250 × 250 m field** (5 m cells, ~10 % tetromino obstacle coverage) for **600 s simulation time**. Obstacles transition through all four required states: *intact* → *burning* → *extinguished* → *burned*.

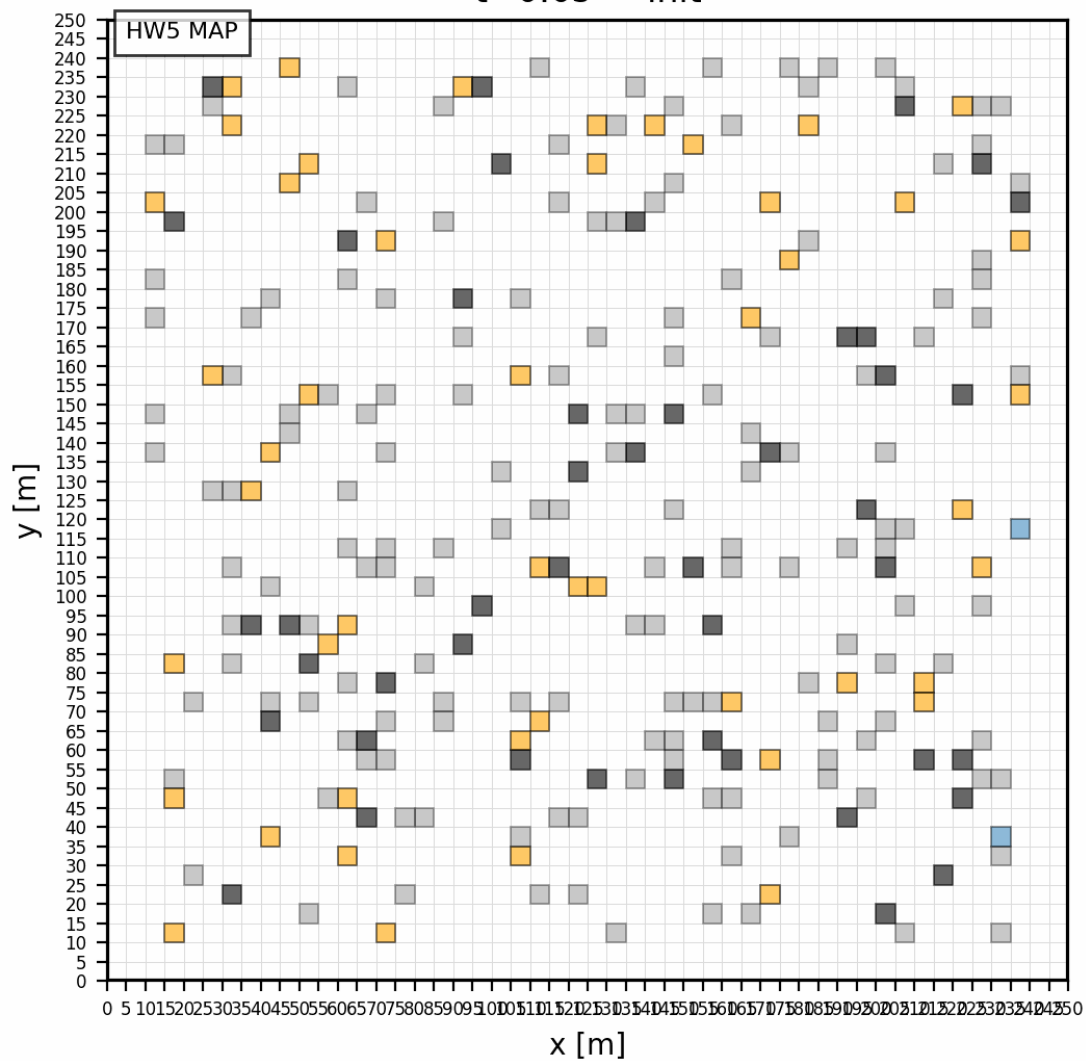
Animations:

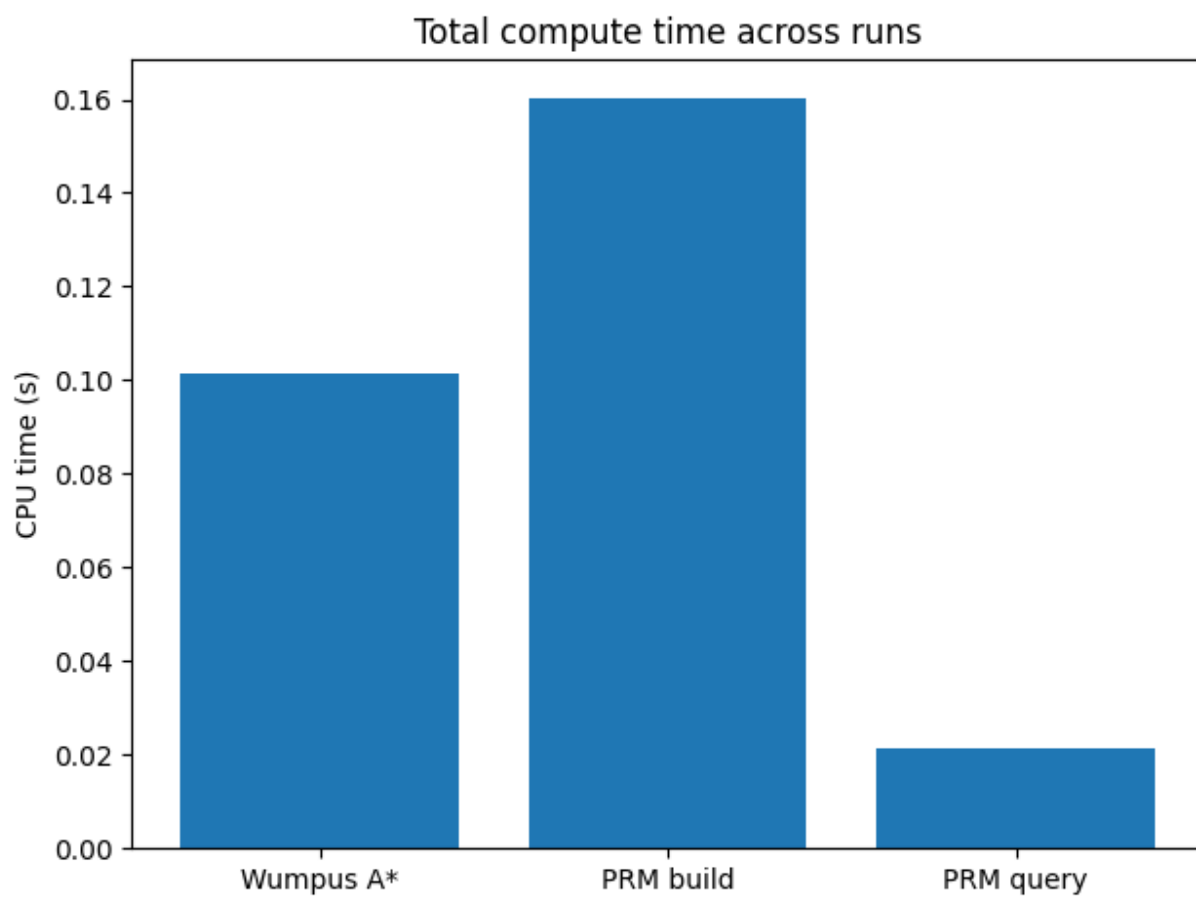


t=0.0s — init



t=0.0s — init





Results Summary:

Run	Wumpus Score	Firetruck Score	Winner
1000	483	34	Wumpus
1001	481	38	Wumpus
1002	482	36	Wumpus
1003	483	34	Wumpus
1004	490	20	Wumpus

Average Wumpus: 483.8 **Average Firetruck:** 32.4 **Champion: Wumpus (5 / 5)**

These results confirm that the discrete A* planner ignites obstacles faster and more aggressively, while the Firetruck PRM is limited by kinematic constraints and extinguish time.

Computation Resources:

Planner Component	Total CPU Time (S)	Relative Cost
Wumpus A*	0.5	36%
PRM build	0.78	56%
PRM query	0.11	8%
Total	1.39	100%

Discussion:

The A* planner for the Wumpus spreads fire aggressively due to deterministic grid traversal, while the PRM firetruck relies on continuous-space sampling and path smoothing to reach and extinguish fires. The PRM build stage is the most computationally expensive, while A* dominates in total runtime effectiveness. Overall, the Wumpus demonstrates greater coverage efficiency, whereas the PRM achieves smoother kinematic feasibility.